Chapter 9

Income Taxes

There are three prominent tax bases: Income, consumption, and property. This chapter presents income taxation and the subsequent chapters cover consumption and property taxes, which are mostly levied at the state and local level. Keep in mind that with respect to U.S. income taxes, households are taxed and not individuals.

Income taxes are one of the most visible and significant forms of taxation for both federal and state governments, typically structured as a progressive tax. They also serve as an important revenue source for local governments, largely through federal and state aid. Changes in federal income tax policies, such as exemptions or deductions, are often automatically implemented at the state level, impacting state and local revenues. Payroll taxes as another key component fund Social Security, Medicare, and unemployment compensation.

Prior to 1913, the federal government raised revenue from taxation of goods, i.e., imported products and excise taxes. At the state and local levels, revenue was raised through poll and property taxes. The 16th Amendment from 1909 states the following:

> *The Congress shall have power to lay and collect taxes on incomes, from whatever source derived, without apportionment among the several States, and without regard to any census or enumeration.*

In 1913, the income tax collection obtained tax returns from less than 1% of the population and the tax rate was 1% (see 16th Amendment to the U.S. Constitution: Federal Income Tax (1913) for additional information). The 2020 distribution of tax return and adjusted gross income is shown in Figure 9.1.

Let us first cover some arguments in favor of and income taxation. On one hand, the income taxation fulfills the tax evaluation criteria of equity since it measures ability to pay. The tax can also be adjusted to the circumstances of the individual household. In addition, the income tax is also adequate with regard to

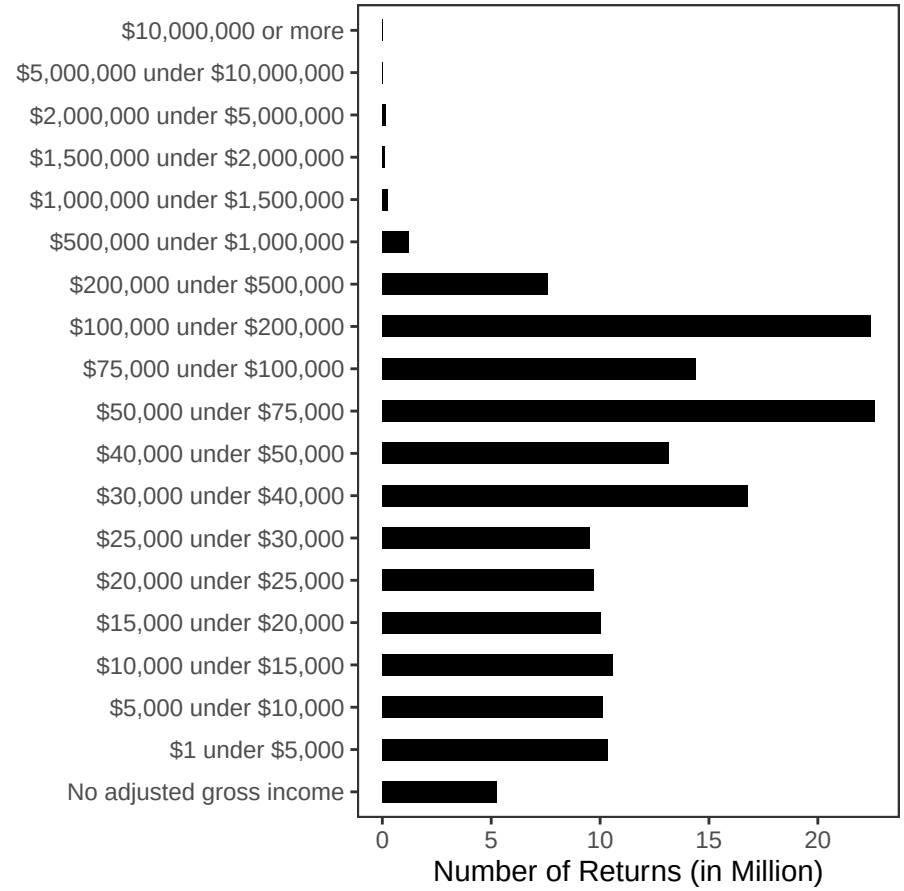


Figure 9.1: Number of tax returns by adjusted gross income (AGI) for 2020. Source: IRS Individual Statistical Tables by Size of Adjusted Gross Income 2020.

tax yield. That is, if households have increasing incomes (i.e., due to economic growth), government revenue increases as well. The tax base is also very broad allowing for higher amounts of revenue at socially acceptable rates. On the other hand, income taxation has some adverse economic effects since it discourages savings and investment of households because those returns are taxed. Some argue that there is also little association between income taxation and the good and services provided by the government. There is also the argument that the individual income tax is complicated and not very transparent.

9.1 Income Definition

The Haig-Simons definition of income states that it is equal to the value of consumption plus the change in net wealth during the year. This is similar to the economic definition of income as the sum of consumption and savings. Sometimes savings are referred to as net savings because the term can also be negative (i.e., representing a decrease in savings).

$$\text{Income} = \text{Consumption} + \Delta\text{Net Wealth}$$

In the ideal case stated by Haig-Simons (1938), taxing should be based on a comprehensive measure of income. However, this is not always achievable. Income taxes currently include wages and salaries, rents, interest income, dividends, realized capital gains. Excluded income sources are fringe benefits (to earned income), gifts, imputed income (e.g., farmer consuming crop), accrued capital gains (i.e., unrealized). In addtion, interest on state and local government debt is excluded. This exclusion allows governments to borrow below market rates (e.g., 5.2% versus 8% if 35% marginal tax rate). To see the differences between the Haig-Simons definition and the current tax system, consider the following cases.

Case 1: Cullen owns stock that increased by $10,000 but he did not sell.

- Haig-Simons definition: Income
- Current tax system: Unrealized capital gains

Case 2: Dena lives in a house she owns and consumes its services

- Haig-Simons definition: Consumption of housing service and hence, part of income
- Current tax system: No tax and thus providing incentives to purchase assets producing non-cash returns to the owner

Case 3: Kenna inherits $50,000 from her aunt

- Haig-Simons definition: Increase in net wealth
- Current tax system: Not income since aunt paid gift/inheritance tax

9.2 Income Tax Components

The process of determining taxable income begins with total (gross) income, which is then reduced by adjustments to calculate Adjusted Gross Income (AGI). From AGI, taxable income is derived by subtracting either itemized or standard deductions, as well as personal exemptions (although these are suspended until 2025). The resulting taxable income is then used to calculate the tax liability based on applicable tax rates and tax brackets. At the end, tax credits are directly reduced from the tax liability. AGI serves as a key measure of an individual's overall tax-bearing capacity, reflecting their financial ability to shoulder tax obligations.

Adjustments: There are a variety of adjustments made to the gross income. First, we have retirement contributions, e.g., Individual Retirement Account (IRA), that are designed to encourage individuals to save for retirement by offering tax advantages. Transfer payments, such as food stamps, are generally not taxed to avoid unnecessary financial transactions. That is, the government wants to avoid providing assistance to low-income households only to tax that same assistance at a later point. However, unemployment benefits are an exceptions since those are fully taxable. Assistance not targeted at low-income households (e.g., Social Security benefits for higher-income individuals) may also be subject to taxation. Other adjustments to taxable income include alimony, expenses related to job-related moves, student loan interest, and self-employed health insurance payments, which are tax-deductible since corporate health benefits are not taxed. Additionally, unrealized capital gains, employer-provided fringe benefits, gifts, and inheritances (to a certain degree) are generally exempt from taxation. Interest received from state and local government bonds is also typically exempt from federal income tax.

Deductions: Certain expenditures that fall outside a household's control may qualify for tax deductions. These include medical and dental expenses exceeding 7.5% of AGI, losses from casualty or theft above 10% of AGI, and state or local income and property taxes. Other expenditures are encouraged by the federal government through tax incentives that reduce their after-tax cost. Charitable contributions and home mortgage interest on first and second homes are examples of such incentivized actions. Additionally, certain job-related expenses can be deducted, such as costs for education, union dues, work uniforms, and occupational taxes. These deductions help lower the financial burden of maintaining or advancing a career.

Personal Exemptions: Personal exemptions, traditionally a flat amount for each person in a household, provided adjustments based on family size, helping to remove low-income families from the tax system by reducing their taxable income. However, personal exemptions have been suspended until 2025. Deductions and exemptions are designed to adjust taxpayers' ability to pay based on their specific circumstances. By tailoring tax obligations to individual situations, these provisions enhance both horizontal equity (ensuring those with similar abilities to pay are taxed similarly) and vertical equity (ensuring those with different abilities to pay are taxed at appropriate rates).

Tax Credits: A tax credit serves as a direct deduction from an individual's tax liability, reducing the amount of tax owed dollar-for-dollar. Tax credits are often used to stimulate private activities and encourage specific behaviors. For example, credits are offered to promote home energy conservation, support child care expenses, and provide financial relief for low-income workers through the Earned Income Tax Credit (EITC). The EITC, in particular, offers a decreasing percentage of earned income as income rises, helping to alleviate the tax burden for those with lower earnings.

Tax Rates: The average tax rate (ATR) is calculated by dividing the total tax

by taxable income. It represents the overall percentage of income paid in taxes and is always lower than the marginal tax rate, which applies only to the last dollar of income earned. The effective tax rate (ETR) is calculated by dividing the total tax by Adjusted Gross Income (AGI), providing a broader measure of tax burden relative to all income sources. The effective tax rate is always below the average tax rate, as it includes deductions and adjustments that reduce taxable income.

9.3 Income Tax Evaluation

Income taxation introduces a potential distortion in the choice between work and leisure, as it taxes income but not leisure. This could discourage work, as individuals may opt for more leisure time rather than earning additional income that will be taxed.

Similarly, the tax system affects the balance between savings and consumption. Savings, particularly interest income, are taxed twice—once when the income is earned and again when the interest is received—while consumption is taxed only once. This structure may encourage people to reduce savings and shift their investments towards real estate or other assets that offer tax advantages over financial investments.

The efficiency of the income tax system also diminishes as marginal tax rates increase. Higher marginal rates can lead to greater efficiency losses, as they create stronger disincentives to earn additional income. This creates a trade-off between achieving greater tax progressiveness and maintaining economic efficiency. Additionally, some deductions and exemptions, while promoting horizontal equity, can further reduce efficiency by distorting individual financial decisions.

9.3.1 New Jersey Millionaires Tax

"*This paper examines the migration response to a millionaire tax in New Jersey, which raised its income tax rate on top earners by 2.6 percentage points to 8.97 percent, one of the highest tax rates in the country. [. . .], we estimate the migration response of millionaires to the rate increase, [. . .]. The results indicate little responsiveness, with semi-elasticities generally below 0.1. Tax-induced migration is estimated to be higher among people of retirement age, people living on investment income rather than wages, and people who work (and pay tax) entirely in-state. The tax is estimated to raise $1 billion per year and modestly reduce income inequality.*"

Young and Varner (2011) Millionaire Migration and State Taxation of Top Incomes: Evidence from a Natural Experiment National Tax Journal 64(2), 255-284.

9.3.2 Equity, Adequacy, and Elasticity

In terms of equity, income taxation is adjusted to account for taxpayers' ability to pay. Horizontal equity is improved by deductions for medical and dental expenses, as well as state and local taxes, ensuring that taxpayers with similar financial circumstances are treated fairly. However, certain provisions, such as the home mortgage interest deduction, can favor homeowners over renters, creating an imbalance in horizontal equity.

Vertical equity, which focuses on the fairness of tax burdens across different income levels, is sometimes compromised by deductions and loopholes that reduce the overall progressiveness of the tax system. These provisions can allow higher-income individuals to reduce their tax liability disproportionately.

In terms of adequacy, income tax is considered a broad-based tax, meaning it has a large tax base and provides substantial revenue, making it a reliable source of funding for government operations.

The elasticity of income tax is tied directly to personal income, meaning tax revenues tend to rise with increasing incomes. However, there is also the potential for bracket creep, where inflation pushes taxpayers into higher tax brackets without an increase in real income, distorting the tax burden.

9.3.3 Bracket Creep and Indexation

In the setup, a taxpayer with a current income of $100,000 falls into the 28% federal income tax bracket. The marginal tax rate increases to 30% for income above $102,000.

Bracket creep occurs when inflation leads to a salary increase, in this case by 3%, raising the taxpayer's income to $103,000. Despite the increase in nominal income, the taxpayer's real purchasing power remains unchanged. However, the higher income pushes them into a higher tax bracket, resulting in a greater tax liability without an actual improvement in their financial situation.

To eliminate bracket creep, tax systems use indexation, which adjusts marginal tax brackets in line with inflation, ensuring that taxpayers do not face higher tax rates simply due to inflationary increases in nominal income.

Feasibility: Income taxation is generally feasible due to its ease of administration for the government. Identifying taxpayers is manageable, and the assessment and collection of taxes are streamlined through payroll deduction (withholding), either by individuals or their employers. The government's primary administrative costs are related to enforcing tax laws and conducting sample audits to ensure compliance.

However, income taxation also imposes significant compliance costs on taxpayers. These costs arise from the effort required to file tax returns, understand tax laws, and keep up with any changes or updates in the tax system.

9.3.4 Marginal Tax Rates, Behavioral Responses, and Government Revenue

One of the central questions in income tax policy is whether higher marginal income tax rates necessarily generate additional government revenue. Historically, the United States imposed substantially higher marginal income tax rates than those observed in recent decades. For example, the highest statutory federal income tax rate reached 70% in 1980 before declining to 50% in 1982 and 28% in 1988. However, the highest statutory marginal rate does not necessarily reflect the average tax rate paid by high-income households. In addition, historical tax rules differed across sources of income, and taxpayers could reduce their taxable income through various deductions, exemptions, and other tax preferences.

The revenue generated by an income tax depends on both the tax rate and the size of the taxable income base:

$$R = t \cdot B$$

where R is tax revenue, t is the tax rate, and B is the taxable income subject to that rate.

If the tax base remains unchanged, increasing the tax rate necessarily increases government revenue. However, the assumption of an unchanged tax base may be unrealistic. Higher marginal tax rates can affect incentives to work, save, and invest. They can also encourage taxpayers to change the timing or composition of their income, increase deductible expenditures, or engage in tax avoidance. Consequently, the tax base may decrease as marginal tax rates increase.

Economists frequently measure the responsiveness of taxable income to changes in tax rates using the elasticity of taxable income with respect to the net-of-tax rate, $1 - t$. The net-of-tax rate represents the fraction of an additional dollar of taxable income retained by the taxpayer.

A simplified relationship between the taxable income base and the tax rate can be expressed as:

$$B(t) = B_0 \left(\frac{1 - t}{1 - t_0} \right)^e$$

where B_0 is the initial tax base, t_0 is the initial tax rate, and e is the elasticity of the tax base with respect to the net-of-tax rate.

Under this model, an increase in the tax rate reduces the net-of-tax return and, for positive values of e, reduces the taxable income base. The magnitude of this response depends on the elasticity.

Substituting this relationship into the government revenue equation gives:

$$R(t) = t \cdot B_0 \left(\frac{1-t}{1-t_0} \right)^e$$

This expression illustrates why increasing a marginal income tax rate does not necessarily result in a proportional increase in revenue. The revenue effect depends on the interaction between the tax rate and the behavioral response of taxpayers. The relationship is also related to the Laffer curve, which illustrates the possibility that revenue eventually declines as tax rates increase.

For a progressive income tax, this simplified model is best interpreted as describing the revenue collected from a particular portion of the tax base, such as income subject to the highest marginal rate. Actual revenue estimates require additional information about the income distribution, tax brackets, deductions, tax credits, and behavioral responses.

It is also important to distinguish between changes in tax revenue caused by economic growth and those caused by changes in tax rates. A growing economy generally increases taxable income and government revenue even when tax rates remain constant. In contrast, an increase in tax rates can change taxpayer behavior and therefore the size of the taxable income base itself.

9.3.5 Effective Marginal Tax Rates and Tax Credit Phaseouts

The statutory marginal income tax rate does not always represent the actual tax burden associated with earning an additional dollar of income. Some government programs provide tax credits or benefits that decline as household income increases. Consequently, earning additional income may simultaneously increase income tax liability and reduce the value of government assistance.

The effective marginal tax rate measures the combined effect of these changes on the household's disposable income.

For example, suppose a household faces a statutory marginal income tax rate of 20%. The household also receives a tax credit that decreases by \$0.15 for every additional dollar of income earned. In this case, the effective marginal tax rate is 35%, since an additional dollar of income increases income taxes by \$0.20 and reduces the tax credit by \$0.15.

This distinction is important when evaluating the economic incentives created by progressive income tax systems. A household with relatively low income may face a higher effective marginal tax rate than a substantially wealthier household, even though the statutory income tax schedule is progressive.

Tax credits and other income-based assistance programs therefore create an important tradeoff between equity, revenue adequacy, and economic efficiency. While these programs can improve the financial circumstances of low-income

households, their phaseout provisions can reduce the financial benefits of earning additional income.

9.4 Income Taxation at the State Level

There are various systems of how income taxes are levied at the state level. In a first case, state income tax can be a percentage of the federal tax liability. In that case, revenue is sensitive to changes in the federal tax code. In a second case, the state tax base is equal to the federal taxable income but uses different tax rates. This system is sensitive to changes in the federal base but not the federal rates. In a third case, the state income tax base equals the federal AGI but uses different deductions, credits, and rates. This system has reduced revenue sensitivity as it relates to federal changes. And finally, state tax base and rates are unrelated to the federal taxes. This results in the state income tax system being completely disconnected from the federal system and associated changes. IN reality many states use a flat tax rate, e.g., 3.23% of Adjusted Gross Income in Indiana.

Income taxation at the federal and state level are characterized by some system of deductibility. Let s and f be the federal and state tax rate, respectively. And let x_s and x_f denote federal and state tax liability, respectively. For the following cases, assume that the taxable income is $M = 60,000$, the federal tax rate is $f = 0.25$, and the state tax rate is $s = 0.05$.

In the case of reciprocal deductibility, we have the following formulas:

- State tax liability
$$x_s = \frac{s - s \cdot f}{1 - s \cdot f} \cdot M$$
- Federal tax liability
$$x_f = \frac{f - s \cdot f}{1 - s \cdot f} \cdot M$$

Given the above example numbers, the state liability is calculated as follows:

$$x_s = \frac{s - s \cdot f}{1 - s \cdot f} \cdot M \Rightarrow \frac{0.05 - 0.05 \cdot 0.25}{1 - 0.05 \cdot 0.25} \cdot 60,000 = 2,279$$

And the federal liability is expressed as follows:

$$x_f = \frac{f - s \cdot f}{1 - s \cdot f} \cdot M \Rightarrow \frac{0.25 - 0.05 \cdot 0.25}{1 - 0.05 \cdot 0.25} \cdot 60,000 = 14,430$$

Thus, the total tax liability is \$16,709 and the average tax rate is $16,709/60,000 = 0.2785$. In the case of deductibility, we have the following for the state tax liability:

$$s \cdot M \Rightarrow 0.05 \cdot 60,000 = 3,000$$

And for the federal tax liability:

$$f \cdot M \Rightarrow 0.25 \cdot 60,000 = 15,000$$

Thus, the total tax liability is \$18,000 and the average tax rate is $18,000/60,000 = 0.3$. In the case of state deductability from the federal tax liability, we have the state liability:

$$s \cdot M \Rightarrow 0.05 \cdot 60,000 = 3,000$$

And for the federal tax liability:

$$f \cdot (1 - s) \cdot M \Rightarrow 0.25 \cdot (1 - 0.05) \cdot 60,000 = 14,250$$

Thus, the total tax liability is \$17,250 and the average tax rate is $17,250/60,000 = 0.2875$.

9.5 Corporate Income Tax

Corporate income tax applies to entities such as C Corporations, while S Corporations are considered pass-through entities, meaning their income is taxed at the individual level rather than the corporate level. The corporate income tax base are the net earnings of incorporated businesses. The tax base includes retained earnings and dividends. The cost of earning income and interest payments on debt are tax deductible.

One of the key advantages of corporate income tax is that it allows for the taxation of out-of-state owners for the services consumed within a particular state. It also ensures that retained earnings, which would not be taxed under personal income tax, are captured within the tax system. This adds to the progressiveness of the overall tax structure, as corporations generally have higher incomes.

However, there are also disadvantages. Corporate income tax can have a negative effect on savings and investments and reduce the after-tax returns potentially discouraging these activities. It can also distort financing decisions, favoring debt financing over equity because interest payments on debt are tax deductible, while equity financing is not. Additionally, there is an issue of double taxation with dividends, as they are taxed once at the corporate level and again when distributed to shareholders.

9.6 Exercises

1. ***State Income Tax*** (**): A hypothetical state taxes income up to \$20,000 at 0% and income above \$20,000 at 10%. Ignore deductions and credits for this question.

a. Derive a piecewise formula for tax liability $T(y)$ as a function of household income y, then write the average tax rate $T(y)/y$ for $y > 0$. Distinguish average and marginal rates, including at the \$20,000 threshold.
b. Can a household have an average tax rate of exactly 7.5%? If yes, determine the corresponding income. Can its average rate equal exactly 10% at any finite positive income? Explain using the formula, rather than a single numerical example.
c. Calculate tax liability, average tax rate, and marginal tax rate at \$10,000, \$20,000, \$40,000, and \$150,000. Plot the average tax rate against income for \$1,000–\$200,000 using a spreadsheet.
d. Ask an AI assistant whether the tax is progressive and request a mathematical justification. Evaluate its answer against your formula and graph. Identify a potentially misleading statement an AI system might make about the 10% tax rate.

2. ***Federal Income Tax Calculation*** (**): Consider the following illustrative 2018 federal marginal income tax schedule for a married couple filing jointly. For this exercise, assume all household income is taxable income (TI) and ignore deductions, credits, payroll taxes, and state taxes. The table shows marginal rates and each rate applies only to the income within that bracket.

TI above (\$)	TI up to (\$)	Marginal rate
0	19,050	10%
19,050	77,400	12%
77,400	165,000	22%
165,000	315,000	24%
315,000	400,000	32%
400,000	600,000	35%
600,000	No upper bound	37%

a. Calculate tax liability, marginal tax rate, average tax rate (tax/taxable income), and after-tax income at household income of \$160,000 and \$175,000. Show the contribution of each occupied bracket to total liability.
b. A friend claims: "The promotion pushes the family into the 24% bracket, so their take-home income falls." Ask an AI assistant to evaluate this claim. Identify any errors in its reasoning, and use your own calculations to state the change in after-tax income.
c. Generate a simple spreadsheet that calculates tax liability for any nonnegative taxable income under this schedule. Test it at \$0, \$19,050, \$165,000, \$160,000, and \$175,000. Briefly explain why testing bracket boundaries matters. If AI helped produce the spreadsheet, describe how you verified it.

3. ***Interest-Bearing Assets and Taxes*** (**): Dena faces a 10% marginal

federal income tax rate and plans to invest $8,000. Cullen faces a 35% rate and plans to invest $15,000. A hypothetical high-quality municipal bond earns 5% interest exempt from federal income tax; an otherwise comparable corporate bond earns 6.5% interest subject to federal income tax. Assume that both bonds have identical risk, maturity, liquidity, and transaction costs, and ignore state and local taxes.

a. For each investor, calculate annual dollar interest after federal income tax on both bonds. Recommend one investment to each investor based strictly on this comparison.
b. Derive the break-even marginal tax rate at which an investor is indifferent between the two bonds. Check your formula numerically for Dena and Cullen.
c. Obtain an AI-generated recommendation for each investor, giving it the facts above. Check whether it correctly applies the exemption and marginal rates. Report one limitation of the AI recommendation, especially if the real-world assumptions differ.
d. Now consider a taxable corporate bond yielding 8% instead of 6.5%. Recalculate the recommendation for each investor and explain why a recommendation can change as market yields change.

4. ***Personal Exemptions and Adjustments*** (**): Compare tax year 2017 and tax year 2018 for a married couple filing jointly with two dependent children. Use the historical federal rules applicable in each year, not current-year tax brackets. For this teaching exercise, the 2017 standard deduction is $12,700 and each household member has a $4,050 personal exemption (four exemptions total). For 2018, the standard deduction is $24,000 and personal exemptions are $0. Assume the household qualifies for the student-loan interest adjustment shown below; no other adjustments apply. Ignore all tax credits (including the child tax credit), alternative minimum tax, phaseouts, and other taxes. Use the information in the table to determine whether itemizing is preferable; for state/local income versus general sales tax, do not deduct both.

Category	Value ($)
Salaries	85,000
Interest income, corporate bonds	2,000
Interest income, municipal bonds	850
State and local income taxes	800
Property taxes	2,130
Home mortgage interest	5,470
Contributions to charities	100
Interest on student loans	500

a. Find and cite a primary source (preferably the IRS) for the 2017 and 2018 married-filing-jointly tax brackets. Ask an AI assistant to locate the schedules, but check the actual numbers against the cited sources.

Note any unreliable citation, incorrect year, or filing-status error.

b. For each year, calculate (1) gross income, (2) adjusted gross income, (3) eligible itemized deductions and standard deduction, (4) personal exemptions, (5) taxable income, (6) federal income tax liability, (7) average tax rate (tax / taxable income), (8) effective tax rate (tax / AGI), and (9) marginal income tax rate. Present the results in a two-column comparison table and show sufficient intermediate calculations to reproduce them.

c. Which year produces lower federal income tax liability under the stated simplifications? Calculate the difference in dollars. Explain why this does not necessarily establish which year's actual tax law made a family better off; identify at least two omitted factors.

d. Ask an AI assistant to independently calculate the two tax liabilities. Independently verify the results using R, a spreadsheet, or manual bracket-by-bracket calculations. Describe at least two checks you performed (for example, treatment of municipal-bond interest, dependent exemptions, deductibility of credit-card interest, or selection between standard and itemized deductions). State whether the AI assistant made any errors; if not, explain how you established that its answer was correct.

5. ***Taxing High Incomes and Tax Revenue*** (***): During the 1970s, the highest federal marginal income tax rate in the United States was 70%. This rate was reduced to 50% in 1982 and eventually to 28% in 1988. Those historical rates raise the question on how much higher government revenue would be if those tax rates were applied today. Assume the federal government is considering increasing the highest marginal income tax rate from 37% to either 50% or 70%. For this exercise, assume that these rates apply only to taxable income exceeding $500,000 per household. All tax rates and rules applying to income below this threshold remain unchanged. The following table describes a hypothetical distribution of household taxable income.

Annual Taxable Income	Number of Households
$750,000	1,237,975
$1,250,000	272,680
$1,750,000	108,913
$3,500,000	157,974
$7,500,000	40,567
$15,000,000	26,576

a. Use AI to investigate the historical development of federal income tax rates for 1970, 1980, and 1990. Identify the highest statutory marginal income tax rates in those years and calculate the average income tax rates actually paid by high-income households, i.e., . Verify the historical information using the Internal Revenue Service (IRS).

Explain why a household facing a 70% marginal tax rate would not necessarily pay 70% of its income in federal income taxes.

b. Use AI to construct a spreadsheet estimating annual federal income tax revenue from the portion of household taxable income exceeding $500,000. Calculate the revenue generated under three alternative marginal tax rates: (1) 37%, (2) 50%, and (3) 70%. Assume that households do not change their behavior in response to higher tax rates. Determine how much additional government revenue would be generated under each proposal, expressed in dollars and as a percentage of the baseline revenue.

c. Higher marginal tax rates may affect labor supply, investment decisions, tax avoidance, and the amount of income reported to the government. Economists frequently analyze these responses using the elasticity of taxable income with respect to the net-of-tax rate. For this exercise, use the following simplified relationship:

$$B(t) = B_0 \cdot \left(\frac{1-t}{1-t_0} \right)^{\gamma}$$

where $B(t)$ is the aggregate taxable income subject to the highest marginal tax rate, B_0 is the corresponding tax base under the initial tax rate, t is the proposed marginal tax rate, $t_0 = 0.37$ is the initial marginal tax rate, and γ measures the responsiveness of this tax base to changes in the net-of-tax rate. This is a simplified behavioral model for the top-bracket tax base, not a complete representation of household taxation. Use AI to calculate the resulting tax revenue under four alternative elasticity assumptions: (1) $\gamma = 0$, (2) $\gamma = 0.2$, (3) $\gamma = 0.5$, (4) $\gamma = 1.0$. For each elasticity, estimate the additional revenue generated by increasing the top marginal tax rate to 50% and 70%. Identify the elasticity at which increasing the rate from 37% to 70% no longer generates additional revenue from this tax base.

d. Prepare a short policy memorandum addressing the following questions: (1) Under what assumptions would returning to a 70% marginal income tax rate generate substantially more revenue? (2) Under what assumptions could increasing the marginal rate actually reduce revenue? (3) What economic behaviors might explain the differences between the static and behavioral estimates? (4) Why would historical tax revenues from the 1970s and 1980s not necessarily provide reliable estimates of revenues under a similar tax policy today? (5) Based on your analysis, would you recommend increasing the highest marginal tax rate? Justify your recommendation.

6. ***Progressive Tax System and Work?*** (***): Progressive income taxes impose higher marginal tax rates on households with higher taxable incomes. However, income tax credits and their phaseouts can create situations in which lower-income households face higher effective marginal tax rates than substantially wealthier households. Consider the following hypothetical

federal income tax system:

Household taxable income	Marginal tax rate
\$0–\$20,000	10%
\$20,000–\$80,000	20%
Above \$80,000	30%

In addition, the government provides a refundable tax credit of \$3,000 to households earning \$25,000 or less. For households with income between \$25,000 and \$45,000, the credit is reduced by 15 cents for every additional dollar earned. The credit is completely eliminated at \$45,000. Assume that household earnings equal taxable income before the credit and that no other taxes, deductions, or benefits apply.

a. Use AI to construct a spreadsheet model calculating the following for households earning \$15,000, \$25,000, \$30,000, \$40,000, \$60,000, and \$100,000 annually: (1) Federal income tax liability before credits, (2) Tax credits received, (3) Net federal tax liability after credits, (4) Average tax rate based on household income, and (5) Disposable income after federal income taxes and credits. Use AI to produce a graph illustrating the relationship between household income and the average tax rate. Explain whether the resulting system should be classified as progressive.
b. The effective marginal tax rate measures how much of an additional dollar of earnings is lost through increased tax payments or reduced government benefits. Using AI, determine the effective marginal tax rate faced by households at different income levels. Pay particular attention to households earning between \$25,000 and \$45,000. Can a household earning \$30,000 face a higher effective marginal tax rate than a household earning \$100,000? Explain your answer. Discuss how the tax credit phaseout might affect incentives to increase working hours or accept a higher-paying job.
c. Suppose policymakers propose reducing the credit phaseout rate from 15% to 10%. The maximum credit remains \$3,000, and the phaseout still begins at \$25,000. Consequently, the credit would now be eliminated at \$55,000 rather than \$45,000. Use AI to compare the original and proposed tax credit structures. Determine which households benefit from the reform and by how much. Assume that household incomes remain unchanged following the reform. Calculate how effective marginal tax rates change. Next, estimate the additional government expenditure associated with the reform using the following hypothetical population:

Household income	Number of households
$15,000	80,000
$30,000	60,000
$40,000	40,000
$70,000	30,000
$110,000	10,000

d. Prepare a policy recommendation addressing the tradeoff between: Vertical equity, incentives to earn additional income, and government revenue adequacy. Explain whether the alternative credit structure provides a better balance between these objectives. Your discussion must consider whether the government's objective should be to minimize marginal tax rates, maximize disposable income for lower-income households, or maintain government revenue. Identify at least one important limitation of the AI-generated analysis.

7. ***Federal Tax Reforms and State Government Revenue*** (***): State governments often link their income tax to the federal income tax. Thus, changes in federal deductions, exemptions, and tax rates can affect state government revenue even when the state legislature does not change its own tax laws. Consider a hypothetical state with the following distribution of households.

			Taxable Income	
Group	Number	Average AGI	Before Reform	After Reform
Low income	150,000	$35,000	$18,000	$21,000
Middle income	80,000	$95,000	$70,000	$78,000
High income	20,000	$350,000	$290,000	$310,000

Suppose a federal income tax reform changes deductions, exemptions, and marginal tax rates. As a consequence, the average federal income tax liability for each household group changes. Before the reform, low-income households paid $1,800 in federal income taxes, middle-income households paid $10,000, and high-income households paid $70,000. After the reform, these amounts decreased to $1,500, $8,800, and $55,000, respectively. For this exercise, assume that state income taxes do not affect the federal tax calculations. The state government is considering three alternative income tax systems:

- System A: The state income tax equals 20% of the household's federal income tax liability.
- System B: The state income tax equals 5% of federal taxable income.
- System C: The state income tax equals 5% of state taxable income, defined as federal AGI minus a state exemption of $40,000. Taxable income cannot be negative.

a. Use AI to construct a spreadsheet model calculating total annual state income tax revenue under each of the three systems. Calculate revenues before and after the federal tax reform. Present your results in a table showing the following: Total state income tax revenue before the reform, (2) Total state income tax revenue after the reform, (3) The absolute change in revenue, and (4) percentage change in revenue. Which state tax system is most sensitive to the federal income tax reform? Explain why.

b. For each state income tax system, calculate the average state income tax rate for low-, middle-, and high-income households. Assess the implications for horizontal and vertical equity. Explain how the same federal tax reform can change the distribution of state income tax burdens depending on the state's tax structure.

c. Suppose that, following the federal reform, 5% of high-income households leave the state. Assume that the average incomes and tax liabilities of all remaining households stay unchanged. Use AI to recalculate state income tax revenue under all three tax systems. How much revenue does the state lose because of taxpayer migration? Discuss whether high-income households are necessarily more responsive to income tax changes than other households. Use the New Jersey Millionaires Tax example discussed in the chapter to evaluate the plausibility of the assumed migration response.

d. You are advising the state's governor, who wants an income tax system that provides (1) adequate and predictable government revenue, (2) progressive distribution of the tax burden, (3) limited exposure to federal income tax reforms, and (4) minimal incentives for high-income households to relocate. Use AI to recommend one of the three state income tax systems. Explain the advantages and disadvantages of your preferred system. Your recommendation must acknowledge that AI-generated calculations do not necessarily capture behavioral responses, administrative costs, or changes in the state's economic conditions.